

Laboratory Experiment No. 3

Objective: To construct and analyze the operation of a 4-bit ripple carry adder, and to examine the propagation delay characteristics inherent to this circuit topology.

Theoretical Background

1. 4-Bit Ripple Carry Adder

A 4-bit ripple carry adder is a combinational arithmetic circuit that computes the sum of two 4-bit binary numbers. It is constructed by connecting four full adder stages in cascade, where the carry output of each stage propagates directly into the carry input of the subsequent stage.

Port Configuration:

- **Inputs:** $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ (two 4-bit operands)
- **Outputs:** $S_3S_2S_1S_0$ (4-bit sum) and C_{out} (carry-out bit)

Operation Principle:

- The least significant full adder (FA1) receives $C_{in} = 0$ as its carry input.
- Each subsequent full adder stage processes its bit-pair along with the carry forwarded from the previous stage.
- The carry signal propagates stage by stage from LSB to MSB — a behaviour termed ripple carry.

2. Propagation Delay Analysis

Propagation delay refers to the time interval between a change in the circuit's input and the corresponding stabilization of the output. In a ripple carry adder, this delay is cumulative because each full adder stage must wait for the carry from the preceding stage before producing its correct outputs.

Why Delay Accumulates:

- No full adder can begin its carry computation until its predecessor has settled.
- The carry signal "ripples" through all stages sequentially, accumulating delay at each step.
- The worst-case path runs from the carry-in of FA1 all the way through to the carry-out of FA4.

Delay Calculation:

Let t_{FA} denote the propagation delay of a single full adder stage. Then:

$$\text{Total Delay (n-bit RCA)} = n \times t_{\text{FA}}$$

$$\text{Worst-case Delay (4-bit RCA)} = 4 \times t_{\text{FA}}$$

This linear delay scaling is the primary performance limitation of the ripple carry adder architecture. For wide operand widths, the accumulated carry delay becomes significant and limits the maximum operating frequency of the circuit.

Logisim Implementation — Circuit and Timing Diagrams

The following figures present the Logisim realization of the 4-bit ripple carry adder and its associated timing waveform demonstrating carry propagation delay:

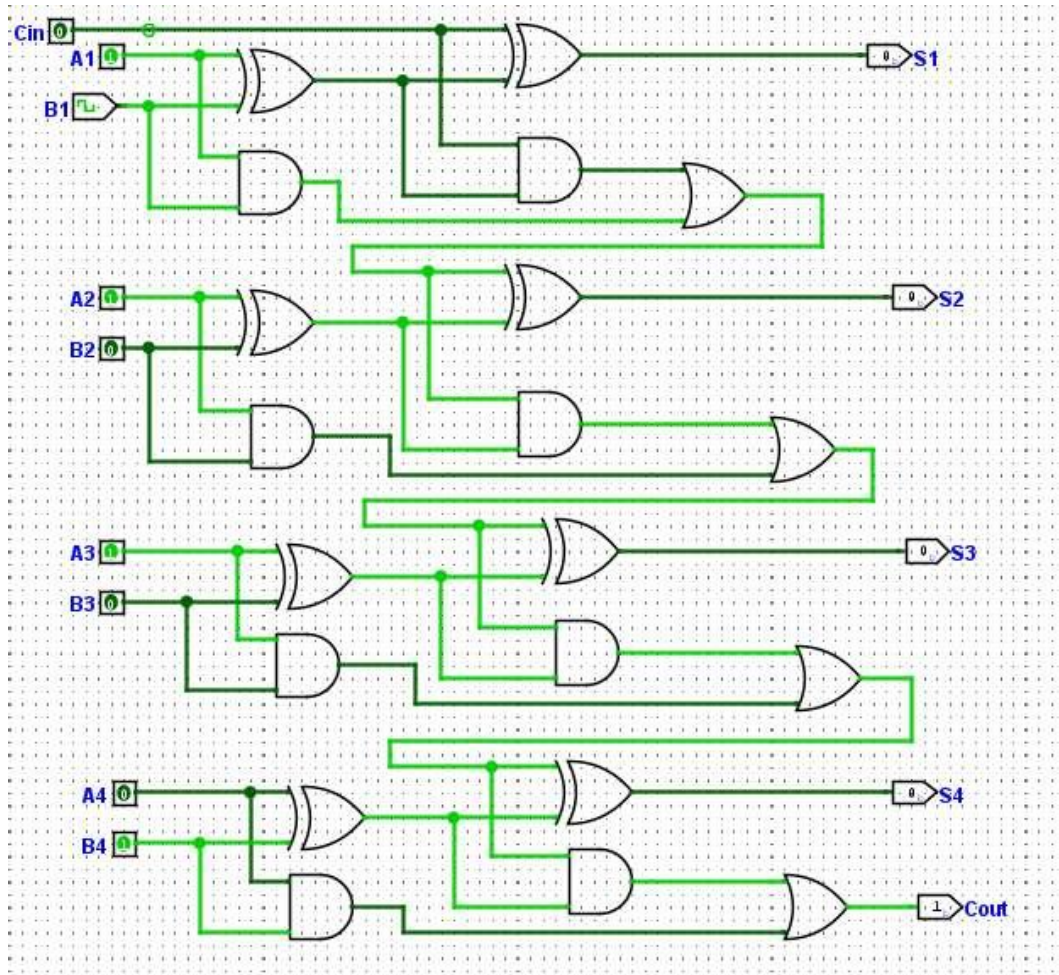


Figure 1: 4-Bit Ripple Carry Adder Circuit in Logisim

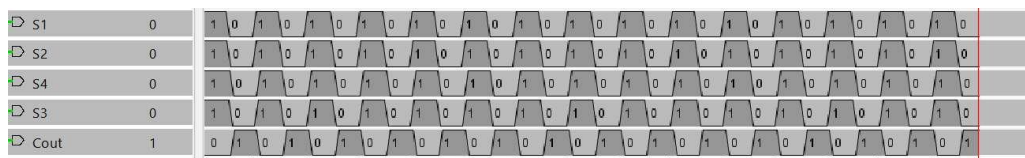


Figure 2: Timing Waveform Showing Carry Propagation Delay (Chronogram View)